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By Amit Katiyar (MCA-JNU)



MATHEMATICS

- The set $(A \Delta B) \cap C$ is not equal to
 - $(A \cap C) \Delta (B \cap C)$
 - $((A \cap B^c) \cap C) \cup ((B \cap A^c) \cap C)$
 - $(A \cap B)^c \cap C$
 - $(A \cap B \cap C)^c \cap ((A \cup B) \cap C)$
- Define a relation $\sim$ on the set $\{1, 2, 3, 4, 5, 6, 7, 8, 9, 10\}$ by $a \sim b$ if $a - 2b$ is divisible by 3. Then
 - $\sim$ is not symmetric, not reflexive not transitive
 - $\sim$ is reflexive but neither symmetric nor transitive
 - $\sim$ is symmetric but neither transitive nor reflexive
 - $\sim$ is transitive but neither reflexive nor symmetric
- Let n be the number of injective function $f: \{1, 2, 3, 4\} \rightarrow \{1, 2, 3, 4, 5, 6, 7, 8\}$ sending an even number to an even number. If $n = 2^a 3^b 5^c$, then $a + b + c$ is
 - 5
 - 6
 - 7
 - 8
- If there are 5 sets represented via a Venn diagram, the number of regions that arise in the Venn diagram equals
 - 22
 - 20
 - 28
 - 32
- Let $R \subset \mathbb{N} \times \mathbb{N}$ which of the following statement is necessarily true
 - If for all each $a \in \mathbb{N}$ the set $R_a := \{b \in \mathbb{N}: (a, b) \in R\}$ has cardinality at most 1 then R represent function $f: \mathbb{N} \rightarrow \mathbb{N}$
 - If for each $b \in \mathbb{N}$ the set $R^b := \{a \in \mathbb{N}: (a, b) \in R\}$ has cardinality atmost 1, then R represent a function $f: \mathbb{N} \rightarrow \mathbb{N}$
 - If for some $a \in \mathbb{N}$ the set R_a is infinite then R_a and R have same cardinality
 - If for all $a \in \mathbb{N}$ the set R_a is infinite then the cardinality of R is larger than that of R_a
- Number of triples of set (A, B, C) with $A, B, C \subseteq \{1 \dots \dots, n\}$ such that $(A \cap B) \subseteq C \subseteq (A \cup B)$ is
 - 5^n
 - 6^n
 - 7^n
 - 8^n
- The number of ordered tuples (p, q, r) in the truth table for which the statements $(\neg p \vee q) \Rightarrow r$ is true is
 - 2
 - 3
 - 4
 - 5
- In a tuition batch of two students, the probability that X will pass the exam is $\frac{2}{5}$ and that of Y is $\frac{3}{4}$. What is the probability that neither X nor Y will pass exam? Assume that outcomes of exam for X and Y are independent of each other
 - $\frac{3}{10}$
 - $\frac{3}{20}$
 - $\frac{1}{5}$
 - $\frac{1}{10}$
- A die is rolled twice independently. What is the Probability that either first throw shows a number less than 4 or second throw shows atleast 4?
 - $\frac{3}{4}$
 - $\frac{2}{3}$
 - $\frac{5}{6}$
 - $\frac{1}{2}$
- Let A_1, A_2, A_3 are three event in sample space with the condition $A_1 \cap A_2 \neq \phi$ then always
 - $P(A_1 \cap A_2 \cap A_3) = P(A_1)P\left(\frac{A_2}{A_1}\right)P\left(\frac{A_3}{(A_1 \cap A_2)}\right)$
 - $P(A_1 \cap A_2 \cap A_3) = P(A_1)P\left(\frac{A_3}{A_2}\right)$
 - $P(A_1 \cap A_2 \cap A_3) = P(A_1)P\left(\frac{A_2}{A_3}\right)$
 - $P(A_1 \cap A_2 \cap A_3) = P(A_1)P\left(\frac{A_1}{A_2}\right)P\left(\frac{A_1}{(A_2 \cap A_3)}\right)$
- The distance (in meter) for seven throw of shotputter are 14.5, 15.2, 16.8, 17.1, 15.9, 16.3, 14.7 calculate sample mean and sample standard deviation. (Rounded to two decimas)
 - 15.79, 1.02
 - 15.79, 0.96
 - 14.79, 1.15
 - 14.79, 0.88





12. Find Mode

Class Interval	No. of Students.
20-25	8
25-30	14
30-35	20
35-40	18
40-45	10
45-50	6

- (a) 32.14 (b) 31.67
(c) 33.75 (d) 34.12

13. An investigator has missed a value while collecting data in an experiment/ Denoting the missing value by x , the observations are: 10, 4, 11, 6, 17, 15, 9, 8, x . What should be the value of x , if he wants mean = median = mode for this data set?

- (a) 9 (b) 10
(c) 11 (d) 12

14. Let X and Y be two independent identically distributed Bernoulli random variables with common probability mass function: $P(X = 1) = \frac{1}{2}$ and $P(X = 0) = \frac{1}{2}$. If $Z = XY$ then distribution of Z is

- (a) Bernoulli with $P(Z = 1) = \frac{1}{2}$ and $P(Z = 0) = \frac{1}{2}$
(b) Bernoulli with $P(Z = 1) = \frac{2}{3}$ and $P(Z = 0) = \frac{1}{3}$
(c) Bernoulli with $P(Z = 1) = \frac{1}{3}$ and $P(Z = 0) = \frac{2}{3}$
(d) Bernoulli with $P(Z = 1) = \frac{1}{4}$ and $P(Z = 0) = \frac{3}{4}$

15. For a given sample the computed values of variance and fourth central movement are 3 and 63 respectively. Then the underlying frequency distribution is classified as

- (a) Normal (b) Mesokurtic
(c) Platykurtic (d) Leptokurtic

16. Let a and b be two positive numbers such that their arithmetic mean is 5 and the harmonic mean is 3.2. Then the numbers are

- (a) $a = 1, b = 9$ (b) $a = 2, b = 8$
(c) $a = 3, b = 7$ (d) $a = 4, b = 6$

17. The roots of the quadratic equation $3x^2 - px + q = 0$ are the 10th and 11th terms of an arithmetic progression with common difference $\frac{3}{2}$. If the sum of the first 11 terms of this arithmetic progression is 88, then $q - 2p$ is

- (a) 464 (b) 474
(c) 484 (d) 494

18. The Coefficient of x^{10} in the expansion of $\left(x^2 + \frac{1}{x}\right)^{12} + \left(x + \frac{1}{x^2}\right)^{12}$ is

- (a) 0 (b) 12
(c) 66 (d) 112

19. The value of the determinant of the following matrix at $x = 2026$ is

$$\begin{vmatrix} x & x+1 & x+3 \\ x+1 & x+3 & x+6 \\ x+3 & x+6 & x+10 \end{vmatrix}$$

- (a) -1 (b) 0
(c) 1 (d) 2026

20. The number of values of θ in the interval $[0, 2\pi]$ for following homogeneous system of equations has non trivial solution is

$$\begin{cases} x + (\sin \theta)y + (\cos \theta)z = 0 \\ x + (\cos \theta)y + (\sin \theta)z = 0 \\ x - (\sin \theta)y - (\cos \theta)z = 0 \end{cases}$$

- (a) 0 (b) 2
(c) 4 (d) 6

21. Let A_k be the arithmetic mean of squares of the first k natural numbers. If n is a natural number such that

$$\sum_{k=1}^n (6A_k - 3k) = 31$$

then, the value of n is

- (a) 1 (b) 2
(c) 3 (d) 4

22. Let a, b, c be non zero number such that $a + b + c \neq 0$ and $4a - 2b + c \neq 0$. If α and β are the roots of quadratic equation $ax^2 + bx + c = 0$. then which of the following equation has roots $\frac{\alpha+2}{\alpha-1}$ and $\frac{\beta+2}{\beta-1}$?

- (a) $(a+b+c)x^2 - (4a+b-2c)x + (4a-2b+c) = 0$
(b) $(a+b+c)x^2 + (4a-2b+c)x + (4a+b-2c) = 0$
(c) $(a+b+c)x^2 - (4a-2b+c)x + (4a+b-2c) = 0$
(d) $(a+b+c)x^2 + (4a+b-2c)x + (4a-2b+c) = 0$





23. If x, y are real number such that $2^{x+\frac{1}{2}} \times 4^{y-\frac{5}{6}} = 3^{x-\frac{1}{2}} \times 9^{y-\frac{1}{3}}$, then which of the following is true?

- (a) $6x + 12y + 7 = 0$
(b) $6x - 12y + 7 = 0$
(c) $6x + 12y - 7 = 0$
(d) $6x - 12y - 7 = 0$

24. Let x, y, z be positive real number such that $2\sqrt{x+y} - 3\sqrt{y+z} = 2$ and $4x - 5y - 9z = 8$.

Then the value of $\sqrt{\frac{20x+38y+18z+1}{9z+9y+2}}$ is.

- (a) 3 (b) 4
(c) 5 (d) 6

25. Which of the following is a value of n if $\left(\sum_{k=1}^n (-1)^{k-1} k\right)^2 - \sum_{k=1}^n (-1)^{k-1} k^2 + 2450 = 0$?

- (a) 98 (b) 99
(c) 100 (d) 101

26. If x, y, z satisfy equation

$$x + y + z = 1,$$

$$4x + 9y + 16z = 25,$$

$$16x + 18y + 256z = 625.$$

simultaneously, then which of the following is true?

- (a) $x = -\frac{36}{15}$ (b) $x = -\frac{15}{36}$
(c) $x = \frac{15}{36}$ (d) $x = \frac{36}{15}$

27. Let $(x_0, y_0) \in \mathbb{Z}^2$ be a point on straight line $8x - 3y = 11$ which is equidistant from the co-ordinate axes. Then the point (x_0, y_0) will lie only in

- (a) I quadrant (b) II quadrant
(c) III quadrant (d) IV quadrant

28. A triangle has vertex on $(1, 2)$ and the mid-points of two sides through it are $(-1, 1)$ and $(2, 3)$. then the area is

- (a) 1 (b) 2
(c) 3 (d) 4

29. Segments of lines $2x + 3y = 1$ and $4x - 3y = 11$ are diameter of a circle of area 153.94 square units. Then the equation of circle with integer radius is.

- (a) $x^2 + y^2 + 4x - 2y - 44 = 0$
(b) $x^2 + y^2 + 4x - 2y + 44 = 0$
(c) $x^2 + y^2 - 4x + 2y - 44 = 0$
(d) $x^2 + y^2 - 4x + 2y + 44 = 0$

30. Which of the following equation represent a common tangent to the parabolas $y = -x^2$ and $y = (x - 2)^2$?

- (a) $y = 4x + 4$ (b) $y = 5x + \frac{25}{4}$
(c) $y = -4x + 4$ (d) $y = -5x + \frac{25}{4}$

31. The eccentricity of ellipse whose centre at origin is $\frac{1}{2}$. If one of its directrices is $x = -4$ then equation of normal to it at $(1, \frac{3}{2})$ is

- (a) $6x - 3y = \frac{3}{2}$ (b) $6x + 3y = \frac{21}{2}$
(c) $4x + 2y = 7$ (d) $4x - 2y = 1$

32. The area of the triangle in the right half plane formed by the lines $x - y = 0, x + y = 0$ and the tangent to the hyperbola $x^2 - y^2 = a^2$, where a is a non-zero number, is

- (a) $\frac{a^2}{2}$ (b) a^2
(c) $2a^2$ (d) $4a^2$

33. Given

$$f(x) = \begin{cases} 1 & \text{if } |x| \leq 1 \\ 0 & \text{if } |x| > 1 \end{cases}, g(x) = \begin{cases} 2 - x^2 & \text{if } |x| \leq 2 \\ 2 & \text{if } |x| > 2 \end{cases}$$

if $h(x) = f[g(x)]$ then the interval in which $h(x) = 1$ for all value of x in that interval is

- (a) $\frac{1}{2} \leq |x| \leq \sqrt{3}c$ (b) $|x| \leq \sqrt{2}$
(c) $1 \leq |x| \leq \sqrt{3}$ (d) $|x| \leq \sqrt{3}$

34. Find the value of $\lim_{x \rightarrow 0} \frac{x^2 + 2 \cos x - 2}{x \sin^3 x}$?

- (a) $\frac{1}{12}$ (b) $\frac{1}{6}$
(c) $\frac{1}{3}$ (d) $\frac{1}{2}$

35. Find the value of $\lim_{x \rightarrow \infty} \left(\frac{\sqrt{x}}{\sqrt{x+\sqrt{x+\sqrt{x}}}} \right) =$

- (a) 0 (b) $\sqrt{2}$
(c) 1 (d) $\sqrt{3}$

36. Find the acute angle at which the curves $y = (x - 2)^2$ and $y = -4 + 6x - x^2$ intersect.

- (a) $\tan^{-1}\left(\frac{4}{7}\right)$ (b) $\tan^{-1}\left(\frac{5}{7}\right)$
(c) $\tan^{-1}\left(\frac{6}{7}\right)$ (d) $\frac{\pi}{4}$

37. $\int_0^{\frac{\pi}{4}} \left(\frac{dx}{\cos^4 x} \right)$

- (a) $\frac{1}{3}$ (b) $\frac{2}{3}$
(c) 1 (d) $\frac{4}{3}$



38. Let f^{-1} denote the inverse of a function f . Let $f: [0, \infty) \rightarrow \mathbb{R}$ defined by

$$f(x) = \frac{x^2 - 1}{x^2 + 1}$$

Then the value of

$$\int_{-1}^1 f^{-1}(y) dy$$

is equal to

- (a) π (b) 2π
(c) $\frac{\pi}{2}$ (d) 0

39. The value of the limit:

$$\lim_{x \rightarrow 0} \frac{|x| \log_e(1 + |\sin 2x|)}{x^2(|x| + 3)} \text{ is}$$

- (a) exists and is equal to $\frac{2}{3}$
(b) exists and is equal to $\frac{1}{3}$
(c) Do not exists
(d) exists and is equal to 0

40. Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = |x + 1|e^{-x^2} \text{ then which of the following of the following statement is true?}$$

- (b) f has point of global maxima in (1,2)
(a) f has point of maxima in $(-2, -1)$
(d) f has point of global minima in (0,1)
(c) f has point of local minima in (0,1)

41. For $a \in \mathbb{R}$, consider real valued function defined on $(-1, 1)$

$$f(x) = \begin{cases} \frac{(1+x)^{\frac{1}{3}} - (1+2x)^{\frac{1}{4}}}{x}, & x \neq 0 \\ a, & x = 0 \end{cases}$$

if f is differentiable at $x = 0$ then value at $a + f'(0)$ is

- (b) $\frac{7}{72}$ (b) $\frac{31}{72}$
(c) $\frac{19}{72}$ (d) $-\frac{1}{6}$

42. Let $f: [0, \infty) \rightarrow \mathbb{R}$ be a function defined by

$$f(x) = \frac{3x^2 + 4x + 1}{x^2 + 3x + 2}. \text{ Then the value of } (f^{-1})'(2) \text{ is}$$

- (a) 5 (b) 25
(c) 15 (d) 0

43. Let $\cos 6x = (a) \cos^6 x + (b) \cos^4 x + (c) \cos^2 x + d$ for any real number x , where a, b, c and d are real constant. Then the value of $a + b + c$ is

- (a) 49 (b) 13
(c) 7 (d) 2

44. Let $x - y \tan 35^\circ = \tan 25^\circ (y + x \tan 35^\circ)$ for some $x, y \in \mathbb{R}$ then which of the following is true.

- (a) $x > y$ (b) $x < y$
(d) $x = y$ (c) no such x, y exists

45. Consider the graph of the function

$$f(x) = 2\cos\left(\frac{x}{2}\right) + 3 \text{ and } g(x) = 4. \text{ The number of points of intersection of the two graphs in interval } [0, 4\pi] \text{ is}$$

- (a) 4 (b) 3
(c) 1 (d) 2

46. If $BC = a, AC = b$ and $AB = c$ are the sides of $\triangle ABC$ and $\angle C \neq \frac{\pi}{2}$. then which of the following is **not** correct?

- (a) $\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$
(b) $\frac{a-b}{a+b} = \frac{\sin A - \sin B}{\sin A + \sin B}$
(c) $\frac{a-b}{a+b} = \cot\left(\frac{A+B}{2}\right) \tan\left(\frac{A-B}{2}\right)$
(d) $\frac{a-b}{a+b} = \frac{\tan\left(\frac{A-B}{2}\right)}{\tan\left(\frac{C}{2}\right)}$

47. An engineer standing at a point P wishes to determine the width of new rectangular pond. She finds the distance to the western-most point A of the Pond from P to be 60 meter, while the distance to the northern-most point B of pond from P is 80 meter. If angle between to lines of sight at P is 60° then width AB in (metres) of pond is

- (AB not parallel to line of north-south)
(a) $20\sqrt{13}$ (b) $13\sqrt{20}$
(c) $10\sqrt{13}$ (d) $13\sqrt{10}$

48. From the top of view point at height of 80 m, the angle of depression of top and bottom of flag standing on same plane are observed to be 30° and 45° , respectively. Find the height (in metres) of the flag.

- (a) $80\left(1 - \frac{1}{\sqrt{3}}\right)$ (b) $80(\sqrt{3} - 1)$
(c) $40\sqrt{3}$ (d) $\frac{80}{\sqrt{3}}$

49. The value of

$$\cos^{-1}\left(\cos\left(-\frac{\pi}{6}\right)\right) + \sin^{-1}\left(\sin\left(\frac{5\pi}{6}\right)\right) \text{ is:}$$

- (a) $\frac{2\pi}{3}$ (b) $\frac{\pi}{3}$
(c) $\frac{5\pi}{3}$ (d) Not Defined



50. The number of solutions of the equation:

$$\tan^{-1}(3x) + \tan^{-1}(2x) = \frac{\pi}{4} \text{ is}$$

- (a) 0 (b) 2
(c) 1 (d) Infinitely many

REASONING

51. Mira's mother-in-law's mother is my grandmother. All my mother's offsprings are unmarried till date. Based on this, which of the following is true?

- (a) Mira is the wife of my uncle's son
(b) Mira is the wife of my aunt's grandson
(c) Mira is the wife of my aunt's son
(d) Mira is my brother's wife

52. Youtube is organizing an event with seven content creators - A, B, C, D, E, F and G, by grouping them into three teams based on - Travel, Beauty and Tech. Each team needs at least 2 content creators, and a creator can be only in a single team. There are certain constraints on the team formations.

- a) A and B refuse to be on the same team.
b) C can be in team Travel or Beauty but not Tech.
c) If D goes to Tech then E also must be in Tech.
d) F must be with either A or B but not both and not in the Travel team
e) G must be in team Travel
f) The Tech team should have exactly 2 members.
g) No team can have more than 3 creators,
h) E must not be in same team as G
i) C and B must be on the same team
j) A cannot be in team Beauty

Who are the members of the Beauty Team?

- (a) B, F, C (b) D, E, C
(c) G, A (d) G, A, F

53. Here you are given one statement and two course of action I and II. Assuming the statements to be true, decide which of the two courses of action most logically follows? Statement: Indian children are very talented but are instead weak in Science and Mathematics,

Courses of Action

- I. Teaching and textbooks are not available in mother language.
II. Education based on experiments in both the subjects is lacking.

- (a) Only I follows
(b) Only II follows
(c) Either I or II follows
(d) Neither I nor II follows

54. The sum of three numbers is 98. If the ratio of the first to the second is 2:3 and that of the second to the third is 5:8, then the second number is

- (a) 20 (b) 30
(c) 48 (d) 58

55. Runs scored by Sachin in a charity match are 10 more than the balls faced by Lara. The number of balls faced by Sachin is 5 less than the runs scored by Lara. d by Lara. Together, they have scored 105 runs and Sachin faced 10 balls less than the balls faced by Lara. How many runs were scored by Sachin?

- (a) 40 (b) 50
(c) 60 (d) 120

56. An examination consists of 160 questions. One mark is given for every correct option. If a 1/4 mark is deducted for each wrong option and a half mark is deducted for each unanswered question, then a student scores 79. If half a mark is deducted for every wrong option and 1/4 mark is deducted for every unanswered question, the person scores 76. Find the number of correct answers he wrote?

- (a) 80 (b) 100
(c) 120 (d) 140

57. Say 10600 is divided into positive quantities A, B, C, and D. Use the following information. to find the value of D: If you remove A, the average of the other 3 numbers is 1000. If you remove B, the average of the other 3 numbers is 3220. If you remove C, the average of the other 3 numbers is 3180.

- (a) 1120 (b) 880
(c) 7600 (d) 1000



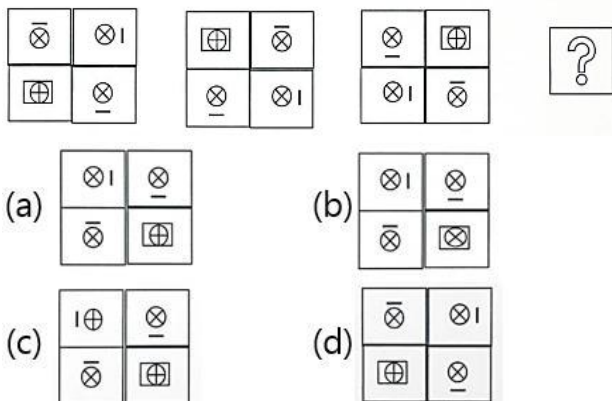


58. A researcher found that 70% of people who own a cat also own a dog, but only 20% of dog owners also own a cat. If there are 1,001 dog owners, how many cat owners are there?
(a) 254 (b) 286
(c) 453 (d) 771
59. Which of the following arguments is a "circular argument"?
(a) If it rains, the grass gets wet. The grass is wet, so it must have rained
(b) Free speech is important because people should be able to say what they want
(c) Most people buy this brand, so it must be the best quality
(d) He is a good doctor because he graduated from a top university
60. In a specific code, "TRUTH" is coded as "7-18-21-20-8" and "FALSE" is coded as "6-1-12-19-5". How would "LOGIC" be coded if the rule changes such that every vowel is shifted by +2 in the alphabet before being converted to its numerical position?
(a) 12-17-7-11-3 (b) 12-15-7-9-3
(c) 12-17-7-9-3 (d) 10-17-5-9-1
61. Six people U, V, W, X, Y, and Z stand in a line. Y is between V and Z. W is to the immediate left of V. X is not at either end. U is to the immediate right of Z. If W is at the far left, who is in the 4th position from the left?
(a) U or V (b) X or Y
(c) 1V or W (d) W or Z
62. Statement 1: All polymers are compounds.
Statement 2: Some compounds are not plastics.
Statement 3: All plastics are synthetic.
Which of the following must be false?
(a) Some polymers are synthetic.
(b) Some synthetic materials are not compounds.
(c) No polymer is a plastic.
(d) All compounds are synthetic.
63. In a school, 70 play only hockey. 100 plays only football. 30 plays both the games. 150 do not play anything at all. What is the percentage (rounded to two decimals) of hockey players to total students in the school?
(a) 46.6 % (b) 20 %
(c) 28.57 % (d) 66.6 %
64. Two statements are given below followed by two conclusions numbered (1) and (2). Which of the given conclusions logically follows from the two given statements? Please disregard commonly known facts.
Statements:
Some professors are doctors.
All the doctors are patients.
Conclusions:
1. Some professors are patients.
2. No doctor is professor.
(a) Only (1) conclusion follows
(b) Only (2) conclusion follows
(c) Neither (1) nor (2) follows
(d) Both (1) and (2) follow
65. **Read the given paragraph. Select a conclusion which can be closely deduced.**
Gig and platform workers need to work for at least 90 days annually with an aggregator to avail of social security benefits, said the final set of rules formulated under the new Code on Social Security (CoSS). In case a worker is engaged with multiple aggregators, the threshold is raised to 120 days, a decision that will affect those working with Swiggy and Zomato or Uber, Ola and Rapido. The rules pave the way for the states to notify their own rules by taking cue from the central ones. Under these latest CoSS rules, an eligible gig and platform worker includes all such workers engaged by the aggregator directly or through an associate, holding or subsidiary company or through a third party. Any income earned from aggregator on a day will be treated as one-day with the platform. For those on multiple platforms, workdays are cumulative. For instance, earning from three aggregators in one calendar day will be counted as three days of engagement.
(a) All gig workers are automatically eligible to access social security benefits
(b) If a gig worker works in three aggregator platforms, they need to work for 120 days in each platform to get the benefits
(c) Aggregator platforms like Blinkit and Flipkart minutes who work on same business models as Swiggy or Rapido are not covered under Code on Social Security
(d) Code of Social Security is given by the central government





66. Find the missing figure in the sequence?



67. Which is the next number in the sequence:

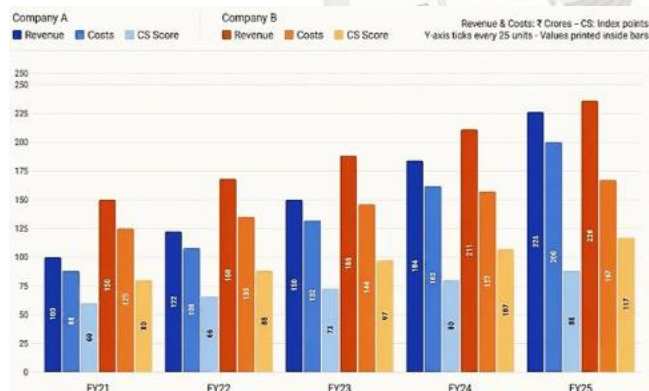
28, 327, 464, 5125, _____?

- (a) 6216 (b) 6126
(c) 6621 (d) 6612

68. If India is coded as JLGEF, then Rome be coded as

- (a) SQPA (b) SMPA
(c) OMPA (d) QQPA

69. The following chart presents data companies A and B on following three Rev Customer Satisfaction (CS) score.



Which of the two companies has higher profit growth rate?

- (a) Company A
(b) Company B
(c) Both companies have same profit growth rate
(d) Information provided is insufficient to answer the question

70. Six files in a directory are labelled P, Q, R, S, T, and U. You are tasked with identifying specific files based on their system attributes:

- (i) Files P, Q, and R are Word Documents (.docx), while S, T and U are Excel Spreadsheets (.xlsx)
(ii) Files Q, R, T, U marked as Read-only, while the others are Editable

(iii) Files P, Q, and S are backed up to the Cloud, while the others are stored only on the Local Disk

Which two files are Read-only, Excel Spreadsheets stored only on the Local Disk?

- (a) R and T
(b) S and U
(c) Q and R
(d) T and U

71. Let W, X, Y, Z be some entities,

Statements:

- a) All Zs are Ys.
b) No Y is a X.
c) Every X is a W.

Conclusions:

- I. Some Ws are Zs.
II. Zs are not Xs.

Then

- (a) Neither conclusion I nor II follows
(b) Only conclusion I follows.
(c) Only conclusion II follows
(d) Both conclusions I and II follow

72. Kartik has three solid objects— a cone, a hemisphere, and a cylinder.

All three have the same base radius and the same height. He completely immerses each solid in a bucket full of water.

What is the ratio of the volumes of the cylinder: cone: hemisphere?

- (a) 1:2:3 (b) 3:1:2
(c) 2:1:3 (d) 1:1:1

73. Consider the sequence:

72, 69, 66,

The numbers continue in the same pattern as long as they remain positive. What will be maximum possible sum of terms of this sequence?

- (a) 900 (b) 897
(c) 882 (d) 903

74. Find the missing Number in the following table?

4	5	3	2	0
7	3	4	4	21
6	4	4	5	22
9	6	5	5	?

- (a) 34 (b) 42



- (c) 44 (d) 45
75. Vikram starts from a point and walks 18 metres towards the West. He then turns left and walks 14 metres. After that, he turns right and walks 11 metres. He again turns right and walks 7 metres. Then he turns left and walks 16 metres. Finally, he turns right and walks 9 metres. In which direction is he facing at the end?

(a) North (b) South
(c) East (d) West

76. Six senior analysts-Arjun, Bhavesh, Charu, Devika, Eshan, and Farah are seated around a circular conference table facing the centre. Each works in a different domain Finance, Marketing, Operations, Analytics, HR, and Strategy. The following information is known:

1. The Finance analyst sits third to the left of Arjun.
2. Bhavesh sits immediately between Devika and the Strategy analyst.
4. Eshan is not an immediate neighbour of Arjun.
3. The Marketing analyst sits opposite the HR analyst.
4. Devika works in Operations.
5. Farah sits second to the right of the Analytics analyst.
7. Arjun does not work in Finance or HR.
8. The Strategy analyst is not Farah.

Who is the Finance analyst?

(a) Arjun (b) Bhavesh
(c) Eshan (d) Farah

77. X,Y,W, Z jointly purchased an office space for Rs 84 lakhs. The amount contributed by Y, W and Z put together is three times of X. The amount contributed by X, W and Z put together is 320% of Y. Further, the amount contributed by W is 20% of X,Y and Z put together. What is the contribution of Z?

(a) 32 lakhs (b) 14 lakhs

(c) 29 lakhs (d) 42 lakhs

78. Suppose the price of liquefied petroleum gas (LPG) increases by 16%, by how much percentage of the consumption of LPG be reduced by the household in order to keep the expenditure on LPG at the same level? (Rounded to two decimals)

(a) 13.79% (b) 15.26%
(c) 16.66% (d) 33.3%

79. There are three products A, B and C and its composition of micronutrients are given below.

Product	Protein%	Carbohydrates%	Fat%
A	20	10	40
B	10	20	30
C	20	15	10

Ranveer is very health conscious and wants to pick a diet that is highest in protein and lowest in fat. From the options given, which diet should he choose?

- (a) 200 grams of B and 200 grams of C
(b) 50 grams of A and 200 grams of B
(c) 350 grams of C
(d) 100 grams of A and 250 grams of C

80. Crossroads, a wellness centre allows a member to spend a maximum of one hour in a meditation hall. Shailaja has a has been in a meditation hall for at least one and a half hours. Which of the following is a valid conclusion if only the above information is used?

- (a) If Shailaja is meditating at Crossroads, she will not be allowed entry the next time
(b) Shailaja is not meditating in the meditating hall at Crossroads
(c) The meditation hall is not spacious to accommodate all the members
(d) If Shailaja is not violating the wellness centre rule, she is not meditating at Crossroads

81. In a class a student ranked 35th from top and 32nd from bottom. How many students are there in the class?

(a) 66 (b) 67





- (c) 78 (d) 23
82. A hand-held gaming device takes X and Y as two input values. These values get updated as $X = XY/2$ and $Y = Y+1$ at each iteration and the game stops when X is greater than or equal to N. For $X=2$, $Y=4$ and $N = 3000$, what would be the final value of X when the game stops?
- (a) 5450 (b) 9450
(c) 12860 (d) 13450
83. Rajan invests an amount of INR 15,860 in the names of his three sons Rohan, Sohan and Mohan in such a way that they get the same interest amount after two, three and four years respectively. If the rate of simple interest is 5% then the ratio of amounts invested among Rohan, Sohan and Mohan will be four years
- (a) 110 : 15 : 20 (b) 22 : 23 : 24
(c) 6 : 4 : 3 (d) 2 : 3 : 4
84. Seven people of a family P1, P2, P3, P4, P5, P6 & P7 go on a picnic in a sports utility vehicle. P4 is the sister of P7, P2 is the mother of P6's wife. P6 is the son-in-law of P3. P5 and P7 are the grandsons of P3. Two people in that group of 7 are fathers; two are brothers; and two are mothers; and one is a sister. What is the relationship between P1 and P4?
- (a) Mother - Daughter (b) Father - Son
(c) Brother - Sister (d) Husband - Wife
85. Five boxes P, Q, R, S, T are stacked one above the other. R is above S but below Q. T is at the bottom. P is just above T. Which box is at the top?
- (a) R (b) Q
(c) S (d) P
86. Pointing to a woman, Arun says
She is the daughter of my father's only son.
How is the woman related to Arun?
- (a) Sister (b) Daughter
(c) Mother (d) Cousin
87. Five years ago, A was three times as old as B. Now, A is twice as old as B. What is A's current age(in years)?
- (a) 20 (b) 15

- (c) 25 (d) 30
88. A train crosses Park Street every forty-five minutes. About 15 minutes ago, one train has crossed Park Street, and the next train will be crossing Park Street by 9.45 am. What is the time now?
- (a) 8.45 am (b) 9.00 am
(c) 9.15 am (d) 9.30 am
89. 10 people went to a restaurant to have a meal as a group. Of these, 9 had already contributed Rs 400 each, while the 10th person had to give the entire additional amount required to honour the restaurant's bill. How much did this person pay if he had to pay Rs 900 more than the group's average?
- (a) Rs 1360 (b) Rs 1350
(c) Rs 1400 (d) Rs 1300
90. The ratio of alcohol to water in two containers, A and B, is 5:3 and 1:3, respectively, with both containers having infinite capacity. Suppose that the aim is to obtain 2.1 litres of liquid, which is composed of equal quantities of alcohol and water. How much liquid should be drawn from A (in litres)?
- (a) 1.4 (b) 1.05
(c) 0.35 (d) 1.1

COMPUTER AWARENESS

91. An embedded computer uses 8-bit 2's complement representation for signed integers. What is the minimum negative integer value which can be represented in this computer?
- (a) 0 (i.e. no negative integers can be represented)
(b) -8
(c) -128
(d) -256
92. A hard drive system has 2 circular disks with a total of 4 surfaces. The disk has 5000 tracks, each with 2000 sectors. How many sectors can be read without the reading head having to make a mechanical movement?
- (a) 10000 (b) 20000





- (c) 4000 (d) 8000
93. A computer has 16GB of RAM. It typically needs to support 100 processes, each of which require an average of 100MB. Will this computer benefit from a Virtual Memory system? Choose the correct option and reasoning below.
- (a) Yes, since Virtual Memory is faster than RAM
(b) No, since total memory requirement is less than total RAM available
(c) Yes, since Virtual Memory provides protection across different processes
(d) No, since Virtual Memory makes the system vulnerable to security attacks
94. Which ONE of the following is a DISADVANTAGE of using dynamically linked library (DLL), compared to using statically linked library?
- (a) Executable file size is larger with DLL
(b) RAM usage is larger with DLL
(c) A program cannot take advantage of bug-fixes in the DLL, long after the program is written
(d) None of the other options
95. Which ONE of the following protocols is used specifically by an email client to read emails from an email server?
- (a) DNS-Domain Name System
(b) ICMP-Internet Control Message Protocol
(c) IMAP-Internet Message Access Protocol
(d) SMTP - Simple Mail Transfer Protocol
96. Which one of the following is correct order of memory types in increasing access speed (Slowest to Fastest)?
- (a) Hard Drive → RAM → Cache → CPU Registers
(b) RAM → Hard Drive → Cache → CPU Registers
(c) Cache → RAM → CPU Registers → Hard Drive
(d) Hard Drive → Cache → RAM → CPU Registers
97. When navigating to a website (Web Browsing), your computer sends a request to a server. If the server's IP address is not known, which system is responsible for "translating" the human-readable URL (e.g. www.google.com) into a machine-readable IP address?
- (a) SMTP (Simple Mail Transfer Protocol)
(b) DNS (Domain Name System)
(c) HTTP (Hypertext Transfer Protocol)

- (d) SSD (Solid State Drive)
98. In the context of "Sending, receiving, and managing emails", what is the primary difference between the POP3 and IMAP protocols?
- (a) POP3 is used for security, while IMAP is used for browsing.
(b) POP3 downloads and usually deletes the email from the server, while IMAP syncs the email across multiple devices.
(c) IMAP can only be used on Linux, while POP3 is for Windows.
(d) There is no difference; they are different names for the same protocol.
99. Which one of the following is decimal equivalent of 8-bit two's complement number 1101 0011?
- (a) -45 (b) -44
(c) -211 (d) +211
100. A CPU uses a 16-bit instruction format. If 4 bits are used for the opcode and the remaining bits specify a single memory address, what is the maximum addressable memory space for this instruction format?
- (a) 4,096 locations (b) 1,024 locations
(c) 65,536 locations (d) 16 locations
101. In a standard machine language instruction, which component identifies the specific operation to be performed, such as addition or data movement?
- (a) Operand (b) Opcode
(c) Register (d) Immediate Address
102. Consider the following table.

I.	Solid State Drive	A.	Volatile main memory that holds data currently in use by the OS and applications
II.	Cache Memory	B.	Long-term, non-volatile storage used for the operating system and large files
III.	Random Access Memory (RAM)	C.	Fastest and smallest storage available for processor computations
IV.	CPU Registers	D.	High-speed buffer memory between the CPU and RAM



		that holds frequently accessed data
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Which of the following options is the correct match between the entries (I-IV) and the descriptions (A-D)?

- (a) I-A, II-C, III-D, IV-B
- (b) I-A, II-B, III-D, IV-C
- (c) I-B, II-D, III-A, IV-C
- (d) I-B, II-C, III-D, IV-A

103. Which component of a web browser is responsible for taking HTML, CSS, and JavaScript code and turning it into the visual page you interact with?

- (a) Javascript Engine
- (b) Rendering Engine
- (c) Network Protocol Stack
- (d) Transport Layer Security Protocol

104. What are 'Cookies' in the context of web browsing?

- (a) The list of all websites that the user has visited in the current browsing session
- (b) Small text files stored on the user's computer to save site preferences and other details
- (c) The list of all websites that the user has visited in the previous browsing session
- (d) Viruses that are used to steal the user's sensitive data

105. Which of the following statements about the fetch-decode-execute cycle in a CPU are correct?

- (a) The Program Counter is incremented after each instruction is fetched so the CPU moves to the next instruction.
- (b) The Control Unit is responsible for fetching instructions and placing them in the Instruction Register.
- (c) The ALU is responsible for decoding instructions and determining which operation to perform.
- (d) Registers are used to permanently store the operating system's files for fast access Identify the CORRECT option.

(a) II and III only

(b) I and II only

(c) I, II, and III only

(d) III and IV only

106. Match the following web-browsing terms with their correct meanings.

Column A: Term		Column B: Meaning	
1.	Web Bro	a.	Converts a human-readable website name into a numerical network address
2.	URL	b.	Program used to request, retrieve, and display web pages
3.	DNS	c.	Standard rules used by browsers and servers to exchange web conter
4.	HTTP	d.	Complete address that identifies the location of a web resource

Which of the following is the CORRECT matching?

(a) 1 → b, 2 → d, 3 → a, 4 → c

(b) 1 → d, 2 → b, 3 → c, 4 → a

(c) 1 → b, 2 → a, 3 → d, 4 → c

(d) 1 → c, 2 → d, 3 → a, 4 → b

107. A system administrator is analysing the software layers of a corporate computer. Which of the following statements correctly distinguish system software from application software?

- I. Device drivers are classified as application software because end-users install them to run devices.
- II. Linux, macOS, and Android are operating systems and are classified as system. software.
- III. The OS kernel manages CPU scheduling, memory allocation, and hardware access on behalf of programs.
- IV. A web browser and a word processor are examples of system software because they help users perform common tasks.

Identify the CORRECT option.



- (a) I and IV only (b) II and III only
(c) I, II, and III only (d) II, III, and IV only

108. Consider the following statements about the range of numbers in a 9-bit 1's complement and 2's complement system.

- I. In 9-bit 1's complement, the range is -255 to +255, and there exist two representation of zero.
II. In 9-bit 2's complement, the range is -256 to 255, and both 1's complement and 2's complement can represent exactly 512 unique values.
III. The maximum positive number representable is +255 in both 1's complement and 2's complement 9-bit systems.

Identify the **CORRECT** option.

- (a) I and II only (b) II and III only
(c) I and III only (d) I, II and III only

109. The multiplication of the following two binary numbers 1100 and 1011 in binary notation is

- (a) 10111 (b) 10000100
(c) 1000100 (d) 1000010

110. The simplified Boolean value of the expression

$(x + y' + z')(x + y' + z)(x + y + z')$ is

- (a) $x' \cdot y + z'$ (b) $x + y'z'$
(c) $x'y + z$ (d) $x \cdot y + z$

GENERAL ENGLISH

111. Choose the sentence(s) that demonstrate correct subject-verb agreement:

- I. Neither the principal investigator nor the associates was available for comment.
II. A series of workshops on research ethics are scheduled for next month.
III. The majority of the manuscript has been revised.
IV. Each of the participants were given an informed consent form.

- (a) I only (b) III only
(c) I and III (d) II and IV

112. How many of the following sentences use the phrasal verbs correctly?

- I. The researchers **carried out** the experiment as planned.
II. The journal **turned down** the paper due to methodological flaws.
III. The student **came across** a rare primary source in the archives.
IV. The professor **called off** the lecture owing to illness.

- (a) 0 (b) 1
(c) 2 (d) 4

113. Match the following:

	Verb		Noun
1.	conduct	A.	a theory
2.	refute	B.	a study
3.	propose	C.	data
4.	analyse	D.	an argument

- (a) 1-B, 2-D, 3-A, 4-C
(b) 1-C, 2-A, 3-D, 4-B
(c) 1-A, 2-C, 3-B, 4-D
(d) 1-D, 2-B, 3-C, 4-A

114. Select the sentence that logically completes the following passage:

"Qualitative research often prioritises depth over breadth. Unlike large-scale surveys that aim for statistical generalisability, ethnographic studies typically focus on a small number of cases. ____"

- (a) Therefore, they usually require larger sample sizes than quantitative studies.
(b) Consequently, their findings are rarely influenced by the researcher's bias.
(c) Hence, they can provide rich, contextualised insights that surveys may miss.
(d) Nevertheless, they are less time-consuming than questionnaire-based research.

115. In the sentence "Brevity is the soul of wit," the word "brevity" means:

- (a) Intelligence (b) Conciseness
(c) Humour (d) Cleverness





116. Select the sentence that correctly expresses the sequence of past events.

- (a) The curator replaces the exhibit after it deteriorated.
- (b) The curator replaced the exhibit after it has deteriorated.
- (c) The curator replaced the exhibit after it had deteriorated.
- (d) The curator replaces the exhibit after it had deteriorated.

117. Identify the underlined phrase in each sentence and match it with its correct type.

	Column A		Column B
A.	The <u>heavy grocery bag</u> slipped from her hand in the kitchen.	(i)	Noun Phrase
B.	He finished his breakfast <u>quite quickly</u> before leaving for office.	(ii)	Adjectival Phrase
C.	The spare keys <u>under the doormat</u> are useful in emergencies.	(iii)	Adverbial Phrase
D.	She felt <u>a bit nervous</u> while waiting for inte results.	(iv)	Prepositional Phrase

- (a) a-(iii), b-(ii), c-(iv), d-(i)
- (b) a-(iv), b-(ii), c-(iii), d-(i)
- (c) a-(iii), b-(iv), c-(ii), d-(i)
- (d) a-(i), b-(iii), c-(iv), d-(ii)

118. The term "Climate change denotes significant, long-term changes in global temperatures and meteorological patterns. Such changes can occur due to natural factors such as changes in the sun's activity and large volcanic eruptions. However, since the Industrial Revolution, anthropogenic activities-specifically the combustion of fossil fuels such as coal, oil and gas-have been the prime causes of climatic shifts. Which of the following is closest in meaning to the underlined word 'anthropogenic' in the text above?

- (a) superinduced
- (b) synthetic
- (c) natural
- (d) human-induced

119. According to a recent newspaper report, an international team of astronomers studying nearly 446,000 galaxies via the Hubble Space Telescope has confirmed that the expansion of the universe is accelerating. By utilising a technique known as "weak gravitational lensing" to observe how light is distorted by dark matter, researchers mapped the distribution of matter and the history of cosmic expansion, determining that an unknown force called "dark energy" has been driving this acceleration for the past two billion years. This study, which involved 1,000 hours of observation and 600 Earth orbits by the Hubble telescope, marks the first time such measurements were successfully conducted using gravitational lensing alone, and it further reinforces the existence of an invisible web of dark matter that constitutes 80% of the universe.

Which of the following BEST summarises the main idea of the text above?

- (a) According to a recent newspaper report, an international team of astronomers have meticulously studied nearly 446,000 galaxies involving 1000 hours of observation and 600 Earth orbits by the Hubble telescope.
- (b) An international team of astronomers has noticed the presence of dark matter in the universe for the first time, according to a newspaper report. They did this by studying nearly 446,000 galaxies through the Hubble Space Telescope.
- (c) According to a recent newspaper report, 'weak gravitational lensing' is the best technique to study dark matter in the universe. An international team of astronomers has arrived at this conclusion after 1000 hours of observation and 600 Earth orbits by the Hubble Space Telescope.
- (d) An international team of astronomers has affirmed that 'dark energy' has been accelerating the expansion of the universe, according to a recent newspaper report. The study used 'weak gravitational lensing' and involved 1000 hours of observation and 600 Earth orbits by the Hubble Space Telescope.



MAARULA CLASSES

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OFFICIAL PAPER



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120. In the case of the recent plane crash, new findings from the analysis of flight data recorder _____ the theory of pilot suicide.

- (a) collaborate (b) collide
(c) collude (d) corroborate

a	b	c	b	b
111.	112.	113.	114.	115.
b	d	a	c	b
116.	117.	118.	119.	120.
c	d	d	d	d

ANSWER KEY

1.	2.	3.	4.	5.
c	c	b	d	c
6.	7.	8.	9.	10.
b	d	b	a	a
11.	12.	13.	14.	15.
a	c	b	d	d
16.	17.	18.	19.	20.
b	b	a	a	c
21.	22.	23.	24.	25.
c	d	c	b	b
26.	27.	28.	29.	30.
d	d	b	c	c
31.	32.	33.	34.	35.
d	b	c	a	c
36.	37.	38.	39.	40.
c	d	a	a	b
41.	42.	43.	44.	45.
a	a	d	a	d
46.	47.	48.	49.	50.
d	a	a	b	c
51.	52.	53.	54.	55.
c	a	b	b	c
56.	57.	58.	59.	60.
b	d	b	b	a
61.	62.	63.	64.	65.
b	d	c	a	d
66.	67.	68.	69.	70.
a	a	b	b	d
71.	72.	73.	74.	75.
c	b	a	a	a
76.	77.	78.	79.	80.
d	c	a	c	d
81.	82.	83.	84.	85.
a	b	c	a	b
86.	87.	88.	89.	90.
b	a	c	c	a
91.	92.	93.	94.	95.
c	d	c	d	c
96.	97.	98.	99.	100.
a	b	b	a	a
101.	102.	103.	104.	105.
b	c	b	b	b
106.	107.	108.	109.	110.

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